ASSESSMENT 2

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SUBJECT:- ADVANCE PYTHON PROGRAMMING

LAB:- L31+L32+L49+L50

Question 1:- (MEDIUM)

7. Reverse Integer

Medium | Topics | Companies

Given a signed 32-bit integer `x`, return `x` *with its digits reversed*. If reversing `x` causes the value to go outside the signed 32-bit integer range `[-2^31, 2^31 - 1]`, then return `0`.

INPUT:- Getting the integer from the user like 123

OUTPUT :- the output is reverse of the input 321

CODE:-

```
a = int(input("Input: "))
d=a
r = []
i = 0
sign=1
if(a>0):
 while a > 0:
    m = a % 10

    if m > 0:
        r.append(m)
```

```
        i = i + 1
    a //= 10
if(a<0):
 a=(-1*a)
 while a > 0:
    m = a % 10

    if m > 0:
        r.append(m)

        i = i + 1
    a //= 10
if d < 0:
    print("-", end="")
    for i in r:
        print(i, end="")
elif d > 0:
    for i in r:
        print(i, end="")
else:
    print("0")
```

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```
a = int(input("Input: "))
d=a
r = []
i = 0
sign=1
if(a>0):
 while a > 0:
    m = a % 10

    if m > 0:
        r.append(m)

        i = i + 1
    a //= 10
if(a<0):
 a=(-1*a)
 while a > 0:
    m = a % 10

    if m > 0:
        r.append(m)

        i = i + 1
    a //= 10
if d < 0:
    print("Output: -", end="")
    for i in r:
        print(i, end="")
elif d > 0:
    print("Output: ",end="")
    for i in r:
        print(i, end="")
else:
    print("Output: 0")
```

```
Input:  -1234
Output: -4321
```

OUTPUT:-

```
Input:  8251
Output: 1528
```

```
Input:  0
Output: 0
```

```
Input:  -1234
Output: -4321
```

JUSTIFICATION:- I used a list because it is simple to store each extracted digit and print them later in the required order

Question 2:-(MEDIUM)

34. Find First and Last Position of Element in Sorted Array

Medium | Topics | Companies

Given an array of integers `nums` sorted in non-decreasing order, find the starting and ending position of a given `target` value.

If `target` is not found in the array, return `[-1, -1]`.

INPUT:- Getting a list elements from the user like [1,2,2,3] and a target number and finding its first and last number if the targets doesn't have first and last position return [-1,-1]

OUTPUT:- the first and last index of the target number

CODE:-

```
num=list(map(int,input("Input: ")[1:-1].split(",")))
target=int(input("Target"))
it=[]
i=0
while i < len(num):
    if target == num[i]:
        it.append(i)
    i += 1
if(len(it)==0):
    {
    print("[-1,-1]")
    }
else:
    {
    print(it)
    }
```

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```
num=list(map(int,input("Input: ")[1:-1].split(",")))
target=int(input("Target"))
it=[]
i=0
while i < len(num):
    if target == num[i]:
        it.append(i)
    i += 1
if(len(it)==0):
    {
    print("[-1,-1]")
    }
else:
    {
    print(it)
    }
```

```
Input:  [1,2,3,4]
Target 4
[3]
```

OUTPUT:-

```
Input:  [5,7,7,8,8,10]
Target 8
[3, 4]
```

```
Input:  [1,2,3,4]
Target 5
[-1,-1]
```

```
Input:  [1,2,3,4]
Target 4
[3]
```

JUSTIFICATION:-

I chose a list because it makes it easy to append and display multiple matching indices

Question3:- (MEDIUM)

24. Swap Nodes in Pairs

Medium Topics Companies

Given a linked list, swap every two adjacent nodes and return its head. You must solve the problem without modifying the values in the list's nodes (i.e., only nodes themselves may be changed.)

INPUT:- getting a list of element from the user and trying to swap the two adjacent elemnt with each other like [1,2,3,4]

OUTPUT:- the swapped linked list gets displayed like [2,1,4,3]

CODE:-

```
class Node:
    def __init__(self, data):
        self.data = data
        self.next = None
arr = list(map(int, input("Input: ")[1:-1].split(",")))
head = None
for x in arr:
    if head is None:
        head = Node(x)
        temp = head
    else:
        temp.next = Node(x)
        temp = temp.next
temp = head
while temp and temp.next:
    temp.data, temp.next.data = temp.next.data, temp.data
    temp = temp.next.next
print("Output: [", end="")
temp = head
while temp:
    print(temp.data, end="")
    if temp.next:
```

```
        print(",", end="")
    temp = temp.next
print("]")
```

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```
class Node:
    def __init__(self, data):
        self.data = data
        self.next = None

arr = list(map(int, input("Input: ")[1:-1].split(",")))

head = None

for x in arr:
    if head is None:
        head = Node(x)
        temp = head
    else:
        temp.next = Node(x)
        temp = temp.next
temp = head
while temp and temp.next:
    temp.data, temp.next.data = temp.next.data, temp.data
    temp = temp.next.next
print("Output: [", end="")
temp = head
while temp:
    print(temp.data, end="")
    if temp.next:
        print(",", end="")
    temp = temp.next
print("]")
```

```
Input:  [1,2,3,4]
Output: [2,1,4,3]
```

OUTPUT:-

```
Input:  [1,2,3,4]
Output: [2,1,4,3]
```

JUSTIFICATION:-

In the question itself they asked to store and treat it as a linked list

Question 4:- (HARD)

4. Median of Two Sorted Arrays

Hard | Topics | Companies

Given two sorted arrays `nums1` and `nums2` of size `m` and `n` respectively, return **the median** of the two sorted arrays.

INPUT:- Getting the two list from the user and find out the median value from that two merged list If it is a even no of elements then the two mid is taken and it is divided by 2

OUTPUT:- Returning the median value

CODE:-

```
num1= list(map(int, input("Input1: ")[1:-1].split(",")))
num2= list(map(int, input("Input2: ")[1:-1].split(",")))
num3=num1+num2
if(len(num3)%2==0):
    q=(len(num3))//2
    print(f"median is{(num[q]+num[q-1])/2}")

if(len(num3)%2!=0):

  q=((len(num3))//2)
  print(f"median is {q+1}")
```

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```
]: num1= list(map(int, input("Input1: ")[1:-1].split(",")))
   num2= list(map(int, input("Input2: ")[1:-1].split(",")))
   num3=num1+num2
   if(len(num3)%2==0):
           q=(len(num3))//2
           print(f"median is{(num[q]+num[q-1])/2}")

   if(len(num3)%2!=0):

       q=((len(num3))//2)
       print(f"median is {q+1}")
```

OUTPUT:-

```
Input1:  [1,2,3]
Input2:  [4,5]
median is 3
```

```
Input1:  [1,2]
Input2:  [3,4]
median is2.5
```

JUSTIFICATION:- using list makes it easy to merge sort and to pick out elements

Question 5:- (HARD)

41. First Missing Positive

Hard | Topics | Companies | Hint

Given an unsorted integer array `nums`. Return the *smallest positive integer* that is *not present* in `nums`.

You must implement an algorithm that runs in `O(n)` time and uses `O(1)` auxiliary space.

INPUT:- getting input as list and searches for the missing +ve number and return the missing number

OUTPUT:- the missing positive integer gets displayed

CODE:-

```
nums = list(map(int, input("Input: ")[1:-1].split(",")))
s = set(nums)
i = 1
while True:
    if i not in s:
        print("Output:", i,"is missing ")
        break
    i += 1
```

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```
nums = list(map(int, input("Input: ")[1:-1].split(",")))

s = set(nums)

i = 1

while True:
    if i not in s:
        print("Output:", i,"is missing ")
        break
    i += 1
```

OUTPUT:-

```
Input:  [1,2,0]
Output: 3 is missing
```

```
Input:  [7,8,9,11,12]
Output: 1 is missing
```

JUSTIFICATION:- I used a set because it stores unique elements